

1: Types of Insurance

1.1 Life Insurance

Definition: A contract where the insurer agrees to pay a specified sum to the beneficiary upon the death of the insured or after a specified period, in exchange for regular premium payments.

Key Features:

- Long-term contract (typically 5-40 years)
- Combines protection with savings/investment
- Nomination and assignment provisions
- Maturity benefits (in some policies)

Types of Life Insurance Policies:

Policy Type	Description	Suitable For
Term Insurance	Pure protection; no maturity benefit; highest coverage at lowest premium	Young individuals, income replacement
Whole Life Insurance	Coverage for entire lifetime; premium paid throughout life	Estate planning, lifelong protection
Endowment Policy	Protection + savings; lump sum paid on maturity or death	Long-term goal planning (retirement, child education)
Money Back Policy	Periodic payouts during policy term; partial survival benefits	Regular liquidity needs
Unit Linked Insurance Plans (ULIPs)	Insurance + market-linked investments; policyholder bears investment risk	Individuals with higher risk appetite
Annuity/Pension Plans	Regular income post-retirement; lump sum converted to periodic payments	Retirement planning

Business Application Example:

A 30-year-old salaried professional with dependent parents purchases a term insurance policy of ₹1 crore for a 25-year term. The annual premium is ₹12,000. If the insured dies during the term, the family receives ₹1 crore. If the insured survives, no maturity benefit is paid – the low premium reflects the pure protection nature.

1.2 Health Insurance

Definition: A contract covering medical expenses incurred due to illness, injury, or hospitalization.

Key Features:

- Indemnity-based (reimburses actual expenses)
- Cashless hospitalization through network hospitals
- Pre-existing disease waiting periods (typically 2-4 years)
- No-claim bonus for claim-free years

Types of Health Insurance:

Type	Description
Individual Health Insurance	Covers a single person up to the sum insured
Family Floater Policy	Sum insured shared among all family members
Senior Citizen Health Plan	Higher premiums, lower waiting periods, specific coverage for age-related ailments
Critical Illness Cover	Lump sum payment on diagnosis of specified illnesses (cancer, heart attack, kidney failure)
Top-up/Super Top-up Plans	Additional coverage above a deductible limit
Group Health Insurance	Employer-provided coverage for employees

Recent Regulatory Updates (IRDAI, 2024-2025):

Regulation	Impact
Cashless authorization within 1 hour	Faster hospital admission
Discharge authorization within 3 hours	Reduced waiting time at discharge
Reimbursement settlement within 15 days	Quicker claim resolution
Interest on delayed claims (prevailing rate + 2%)	Insurer accountability

Business Application Example:

A family of four purchases a family floater health policy with ₹5 lakh sum insured. The father is hospitalized for cardiac surgery with a bill of ₹3.5 lakh. Using the insurer's network hospital, cashless approval is obtained within 1 hour as per IRDAI guidelines, and the family pays nothing out of pocket.

1.3 Fire Insurance

Definition: A contract indemnifying the insured against loss or damage to property caused by fire, lightning, or explosion.

Key Features:

- Principle of Indemnity/cover applies
- Standard Fire and Special Perils Policy (SFSP)
- Underinsurance leads to average clause application

Liabilities Covered (Standard Fire Policy):

Covered Perils	Typically Excluded Perils
Fire (actual ignition)	War and nuclear risks
Lightning	Earthquakes (unless add-on)
Explosion/Implosion	Floods, cyclones (add-on available)
Aircraft damage	Willful negligence
Riot, strike, malicious damage	Spontaneous combustion
Storm, cyclone, typhoon, tempest	Electrical short-circuit (limited coverage)

Add-on Covers Available:

- Flood, cyclone, inundation
- Earthquake (fire and shock)
- Landslide, rockslide
- Terrorism
- Spoilage of stock

Business Application Example:

A textile manufacturing unit with inventory worth ₹2 crore insures its factory under a fire policy. A short circuit causes a fire destroying ₹50 lakh worth of goods. The insurer assesses the loss and indemnifies the business for the actual loss, subject to the sum insured.

Example

Particulars	Value
Actual value of building	₹10,00,000
Sum insured (underinsured)	₹5,00,000
Loss due to fire	₹4,00,000

Without Average Clause: Claim = ₹4,00,000

With Average Clause Applied:

$$\text{Claim} = (5,00,000 / 10,00,000) \times 4,00,000$$

$$\text{Claim} = 0.5 \times 4,00,000 = ₹2,00,000$$

The policyholder bears the remaining ₹2,00,000 as **underinsurance penalty**.

1.4 Motor Insurance

Definition: Insurance covering loss or damage to vehicles and third-party liability arising from their use.

Two Components:

Component	Coverage	Mandatory?
Third-Party (TP) Liability	Legal liability for death/injury of third person or damage to third-party property	Yes (under Motor Vehicles Act, 1988)
Own Damage (OD)	Damage to insured's own vehicle due to accident, theft, fire, natural calamities	Optional

Types of Motor Insurance Policies:

Policy Type	Coverage
Third-Party Only	Only TP liability; no coverage for own vehicle
Comprehensive Policy	TP + OD + personal accident cover for owner-driver
Standalone Own Damage	Only OD coverage (requires separate TP policy)

New Features (IRDAI Master Circular 2024):

- "Pay as you drive" / "Pay as you go" options as first choice
- No claim shall be rejected for want of documents
- No contribution clause in case of multiple policies

Business Application Example:

A car owner purchases a comprehensive motor insurance policy. The car is damaged in an accident with repair costs of ₹25,000. The insurer settles the claim, deducting the applicable deductible (e.g., ₹1,000). If the same car damages a third-party vehicle, the TP liability section covers the third-party's repair costs.

2: Underwriting

2.1 Definition and Purpose

Underwriting is the process by which an insurer evaluates the risk associated with a proposed insurance applicant and determines:

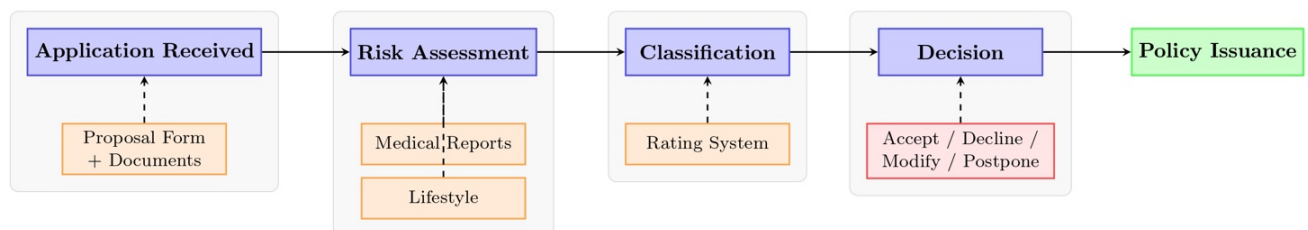
- Whether to accept the risk
- The terms and conditions of coverage
- The premium to be charged

Purpose of Underwriting:

1. Prevent adverse selection (only high-risk individuals seeking insurance)
2. Ensure equitable pricing across risk classes
3. Maintain insurer solvency
4. Comply with regulatory requirements

2.2 Underwriting Process

Insurance Underwriting Workflow



2.3 Factors Considered in Underwriting

Insurance Type	Underwriting Factors
Life Insurance	Age, gender, medical history, family history, smoking/alcohol, occupation, hobbies (adventure sports), income
Health Insurance	Pre-existing diseases, BMI, lifestyle, family medical history, age
Fire Insurance	Building construction type, occupancy, fire safety equipment, location, surrounding hazards
Motor Insurance	Vehicle age, make/model, driver's age and experience, claim history, usage (commercial/personal), geographical area

2.4 Underwriting Decisions

Decision	Description
Standard Acceptance	Normal premium, standard terms
Substandard/Accept with Loading	Higher premium due to increased risk
Accept with Exclusions	Specific conditions/risks excluded
Postponement	Decision deferred (e.g., pending medical investigation)
Declinature	Risk unacceptable (e.g., terminal illness, hazardous occupation without specialized policy)

2.5 Regulatory Requirements for Underwriting

- Required documents to be called at the time of underwriting the proposal
- No claim shall be rejected for want of documents
- Underwriting must consider suitability and affordability of customers
- No improper underwriting practices or unprincipled rate cutting

Business Application Example:

A 55-year-old applicant with controlled diabetes and hypertension applies for a health insurance policy. The underwriter reviews medical reports and recommends a 25% premium loading and a 2-year waiting period for diabetes-related claims. The applicant accepts these modified terms.

3: Claims Management

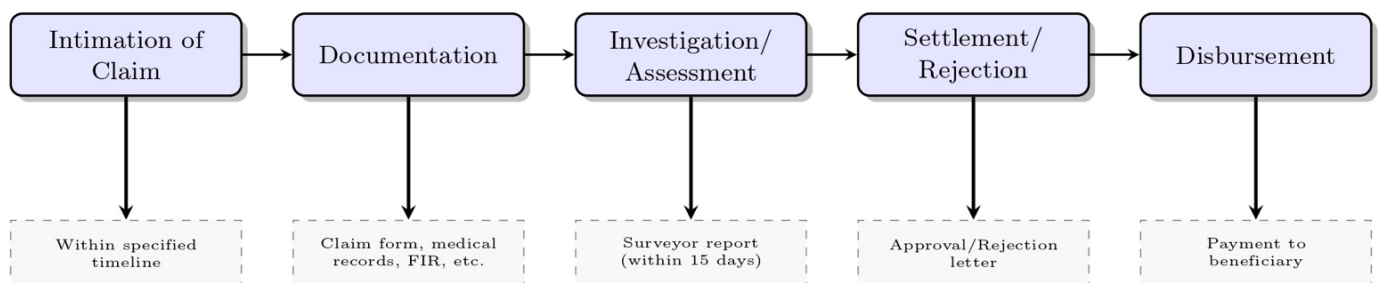
3.1 Definition and Objectives

Claims Management refers to the process of receiving, investigating, evaluating, and settling claims made by policyholders.

Objectives:

- Fair and prompt settlement of genuine claims
- Detection and prevention of fraudulent claims
- Compliance with regulatory timelines
- Customer satisfaction and retention

3.2 Claims Process Flow



3.3 Claims Settlement Timelines (IRDAI Regulations)

Insurance Type	Claim Type	Timeline
Life Insurance	Death requiring investigation	45 days
Life Insurance	Death without investigation	15 days
Health Insurance	Cashless hospitalization authorization	1 hour
Health Insurance	Discharge authorization	3 hours
Health Insurance	Reimbursement claims	15 days
General Insurance	Surveyor appointment	Within 24 hours of claim reporting
General Insurance	Surveyor report submission	15 days
General Insurance	Claim settlement/rejection after report	7 days

3.4 Key Claim Management Principles (IRDAI Master Circular 2024)

Principle	Implication
No claim rejection for want of documents	Documents to be collected at underwriting stage
Interest on delayed claims	Prevailing interest rate + 2%
Salvage disposal is insurer's responsibility	Policyholder receives full claim amount
Random, automated surveyor allocation	No human intervention; fair distribution
No contribution clause	In case of multiple policies, no pro-rata reduction

3.5 Surveyors and Loss Assessors

Role: Independent professionals who assess the extent of loss or damage for claims.

Regulatory Requirements:

- Work allocation on random basis through tech-based solution
- Surveyor to be appointed within 24 hours of claim reporting
- Report to be submitted within 15 days

Types of Surveyors:

- Licensed by IRDAI
- Categories based on claim amount limits
- Specializations (fire, engineering, marine, motor, etc.)

Business Application Example:

A warehouse insured under a fire policy suffers damage due to a short circuit. The policyholder intimates the claim. Within 24 hours, the insurer's system randomly assigns a licensed surveyor. The surveyor inspects the site within 3 days, assesses the loss at ₹12 lakh, and submits the report on day 12. The insurer settles the claim on day 18 (7 days after report receipt), well within regulatory timelines.

4: IRDAI Regulations

4.1 Insurance Regulatory and Development Authority of India (IRDAI)

Establishment: 1999 (IRDA Act, 1999)

Headquarters: Hyderabad

Mission: To protect policyholders' interests and regulate the insurance industry for orderly growth

4.2 Key Functions of IRDAI

Function	Description
Licensing	Granting certificates of registration to insurers, agents, brokers, surveyors
Product Approval	Approving insurance products (move towards use-and-file system)
Tariff Regulation	De-tariffed in most segments; rates determined by insurers
Investment Regulation	Prescribing investment patterns for insurers
Solvency Monitoring	Ensuring insurers maintain required solvency margin
Grievance Redressal	Insurance Ombudsman mechanism
Claim Settlement Oversight	Timelines, fairness, transparency

4.3 Major Regulatory Reforms (2024-2025)

Reform	Impact
Shorter claim timelines	Faster settlements; accountability through interest penalties
Ban on premium collection before policy issuance	Prevents charging for policies not issued
Enhanced free-look period (15→30 days)	More time to review policy terms
Mandatory need analysis for savings/annuity products	Prevents mis-selling; ensures product suitability
Customer Information Sheet (CIS)	Clear disclosure of coverage, exclusions, claim process
Policy cancellation rights	Customer can cancel anytime; insurer only on fraud grounds
No claim rejection for want of documents	Burden of document collection on insurer at underwriting stage

4.4 Protection of Policyholders' Interests Regulations, 2024 (Exposure Draft)

Key Provisions:

- Prohibition on mis-selling and unfair trade practices
- Mandatory disclosure of all material information
- Grievance redressal mechanism with specified timelines
- Free-look period for cancellation
- Nomination and assignment provisions

4.5 Insurance Ombudsman

Establishment: 1998 (Redressal of public grievances against insurance companies)

Jurisdiction:

- Claims disputes (repudiation, delay, quantum)
- Premium disputes
- Policy terms interpretation

Process:

1. Complaint filed within 1 year of insurer's final decision
2. Ombudsman attempts conciliation
3. Award binding on insurer if accepted by complainant
4. Insurers can now appeal ombudsman decisions within 30 days (new provision)

Penalty for Delay: ₹5,000 per day for non-compliance with ombudsman ruling or appeal timeline.

5: Risk Assessment Frameworks

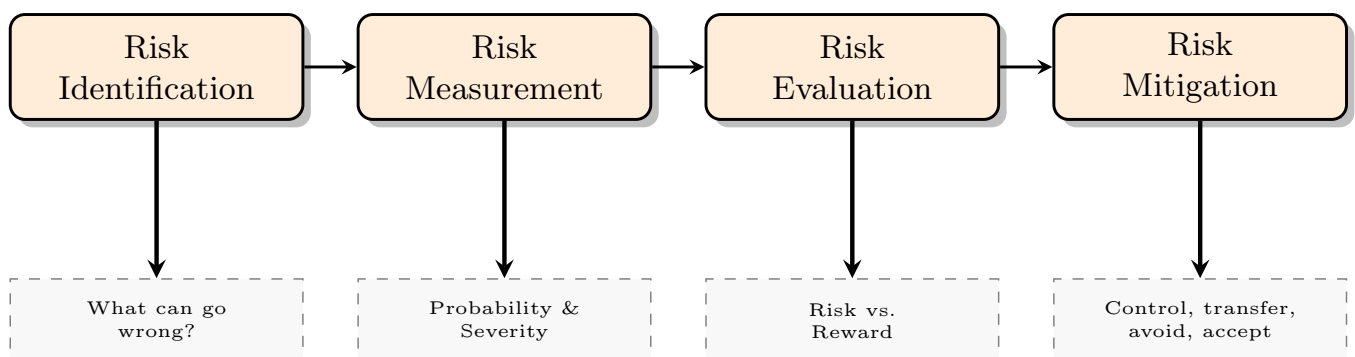
5.1 Definition of Risk in Insurance Context

Risk is the uncertainty concerning the occurrence of a loss. In insurance, risk is the possibility of an adverse deviation from expected outcomes.

5.2 Classification of Risks

Classification	Types
By Nature	Pure risk (loss/no loss) vs. Speculative risk (gain/loss/no loss)
By Frequency	High frequency/low severity; Low frequency/high severity
By Predictability	Insurable (predictable) vs. Uninsurable (unpredictable)
By Source	Financial, operational, strategic, hazard

5.3 Risk Assessment Framework – Key Components



5.4 Enterprise Risk Management (ERM) Framework for Insurers

Component	Description
Risk Identification	Systematic identification of all material risks
Risk Measurement	Quantification using actuarial models
Risk Appetite Statement	Board-approved risk tolerance levels
Risk Limits	Delegated authority and exposure limits
Risk Monitoring	Regular reporting and escalation
Stress Testing	Scenario analysis for extreme events

5.5 Types of Risks for Insurance Companies

Risk Type	Description
Underwriting Risk	Losses due to inadequate premiums or unexpected claims
Credit Risk	Default by reinsurers or other counterparties
Market Risk	Investment losses due to market movements
Liquidity Risk	Inability to meet claim obligations when due
Operational Risk	Losses from failed processes, systems, or people
Regulatory Risk	Non-compliance with IRDAI regulations
Reputational Risk	Damage to brand and customer trust

5.6 Solvency and Capital Adequacy

Solvency (Creditworthiness) Margin: The excess of assets over liabilities that an insurer must maintain to absorb unexpected losses.

Regulatory Requirements:

- Life insurers: 150% of required solvency margin
- General insurers: 150% of required solvency margin

Risk-Based Capital (RBC) Approach:

- Capital requirements linked to risk profile
 - Factors considered: asset risk, underwriting risk, operational risk
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6: Actuarial Basics

6.1 Definition of Actuarial Science

Actuarial Science is the discipline that applies mathematical and statistical methods to assess risk in insurance, finance, and other industries.

"How much should an insurance company charge a customer to cover a specific risk?"

Role of an Actuary:

- Pricing insurance products
- Calculating reserves (liabilities)
- Assessing solvency
- Performing valuation
- Certifying financial statements

The Simple Analogy: A Lemonade Stand Bet

Imagine you run a lemonade stand. There is a 10% chance that rain will ruin your stock worth ₹1000.

- **Expected loss** = $10\% \times ₹1000 = ₹100$

Now, how much should you charge a friend who promises to pay you if it rains?

Principle	Description	Premium
Pure premium	Just the expected loss	₹100
Fair premium	Expected loss + expenses + profit margin	₹100 + ₹20 = ₹120
Risk-loaded premium	Extra charge for uncertainty/variability	₹100 + ₹20 + ₹30 = ₹150

Why Do We Need Different Premium Principles?

Because not all risks are the same.

Risk Type	Example	Challenge
Simple risk	Life insurance (mortality tables)	Easy to predict
Complex risk	Cyber-attack or earthquake	Hard to predict; high uncertainty
Large risk	Oil rig explosion	Rare but catastrophic

Different premium principles handle these situations differently.

Major Premium Principles:

1. Expected Value Principle

"Premium = Expected loss + Safety loading"

Term	Meaning
Expected loss	Average loss you expect to pay (probability × loss amount)
Safety loading	Extra percentage added for uncertainty

Example:

- Expected loss = ₹1000
- Safety loading = 10% of expected loss = ₹100
- **Premium = ₹1100**

Good for: Simple, predictable risks

Bad for: Highly uncertain risks (the 10% loading may not be enough)

2. Variance Principle

"Premium = Expected loss + (Risk factor × Variance of loss)"

Term	Meaning
Variance	How much the loss can swing up and down (volatility)
Risk factor	A number chosen by the insurer (e.g., 0.5)

Example:

- Risk A (stable): Expected loss = ₹1000, Variance = 100 → Premium = 1000 + (0.5 × 100) = ₹1050
- Risk B (volatile): Expected loss = ₹1000, Variance = 1000 → Premium = 1000 + (0.5 × 1000) = ₹1500

Good for: Comparing two risks with same expected loss but different volatility

3. Standard Deviation Principle

"Premium = Expected loss + (Risk factor × Standard deviation)"

Standard deviation = square root of variance (easier to understand in original units).

Example:

- Risk A: Expected loss = ₹1000, Standard deviation = ₹100 → Premium = $1000 + (0.5 \times 100) = ₹1050$
- Risk B: Expected loss = ₹1000, Standard deviation = ₹300 → Premium = $1000 + (0.5 \times 300) = ₹1150$

Good for: Adding a premium loading that is in the same unit as the loss (rupees, dollars)

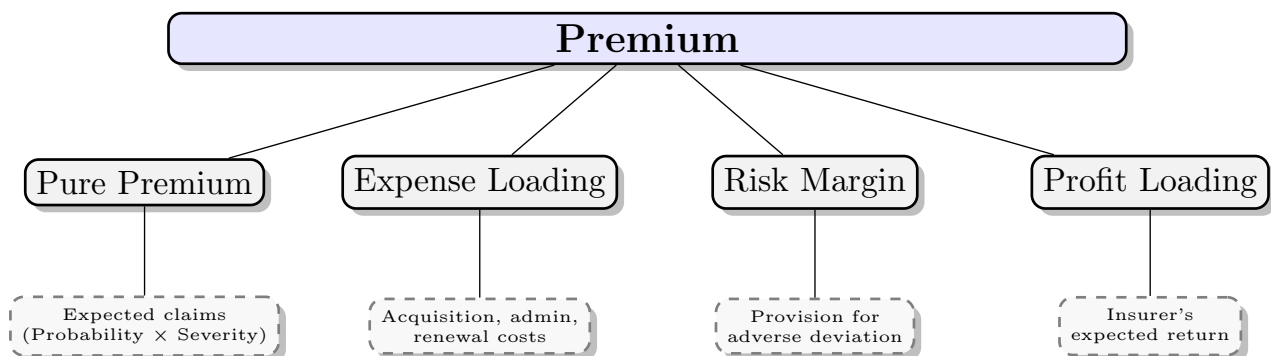
6.2 Key Actuarial Concepts

Concept	Description
Mortality Table (Life Table)	Statistical table showing expected deaths at each age
Morbidity Table	Statistical table showing incidence of illness/disease
Life Expectancy	Average number of years a person is expected to live
Premium	Price of insurance coverage
Reserve	Liability set aside to meet future claims
Discounting	Calculating present value of future payments

6.3 Premium Calculation

Components of Premium:

Premium = Pure Premium + Expense Loading + Risk Margin + Profit Loading



Where:

- Pure Premium = Expected claims (Probability × Severity)
- Expense Loading = Acquisition, administration, renewal costs
- Risk Margin = Provision for adverse deviation
- Profit Loading = Insurer's expected return

6.4 Risk Measures in Actuarial Science

Risk Measure	Description	Formula (Simplified)
Value at Risk (VaR)	Maximum loss at given confidence level over specified period <i>"I am X% certain that I will not lose more than ₹Y over Z days."</i>	$P(\text{Loss} > \text{VaR}) = 1 - \text{Confidence Level}$
Conditional Tail Expectation (CTE)	Expected loss given loss exceeds VaR <i>"If the loss exceeds the VaR threshold, what is the AVERAGE loss I should expect?"</i>	$\text{CTE} = E[\text{Loss} \mid \text{Loss} > \text{VaR}]$
Expected Shortfall	Average loss in worst-case scenarios	Alternative name for CTE

6.5 Actuarial Valuation

Purpose:

- Determine liabilities (reserves) for policyholders
- Assess solvency position
- Calculate surplus for distribution (bonus to policyholders)

Valuation Methods:

Method	Application
Gross Premium Method	Uses actual premiums charged
Net Premium Method	Uses net premiums (excluding expenses)
Prospective Method	Present value of future benefits minus future premiums
Retrospective Method	Accumulated past premiums minus past claims

6.6 Regulatory Role of Appointed Actuary

Responsibilities under IRDAI Regulations:

- Certify asset-liability position
- Provide actuarial report and abstract for life business
- Advise on pricing and product design
- Ensure solvency compliance
- Sign off on valuation reports

Business Application Example:

A life insurance company launches a new term insurance product. The actuary calculates the pure premium using mortality tables (e.g., Indian Assured Lives Mortality table). Expenses (agent commission, administration) are added. A risk margin (say 10%) is included for adverse deviation. The final premium is filed with IRDAI. The same actuary will annually value the reserves for all in-force policies and certify the company's solvency position.

Summary:

Comparison of Insurance Types

Feature	Life	Health	Fire	Motor
Nature	Long-term	Annual/ renewable	Annual	Annual
Principle	Protection + Savings	Indemnity	Indemnity	Indemnity + Liability
Risk Measured	Mortality	Morbidity	Property damage	Accident, theft, liability
Key Regulation	IRDAI (Life)	IRDAI (Health)	IRDAI (General)	IRDAI + MV Act
Claim Type	Death/Maturity	Hospitalization	Property loss	Own damage/TP

Key IRDAI Timelines (2024-2025)

Activity	Timeline
Health cashless authorization	1 hour
Health discharge authorization	3 hours
Health reimbursement claim	15 days
Life claim (no investigation)	15 days
Life claim (investigation required)	45 days
Surveyor appointment	24 hours
Surveyor report	15 days
Claim settlement after report	7 days
Free-look period	30 days

Actuarial Formulas Reference

Concept	Formula
Expected Loss	$\Sigma (\text{Probability of event} \times \text{Severity of event})$
Pure Premium	Expected Loss
VaR at $\alpha\%$	$P(\text{Loss} > \text{VaR}) = 1 - \alpha$
CTE	$E[\text{Loss} \mid \text{Loss} > \text{VaR}]$
